Loan Approval Prediction Using Supervised Machine Learning: A Classification Approach to Automating Preliminary Credit Assessment

Data Mining Group Project

Project Developers

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1. Background of the Problem and Problem Statement

Lending is one of the main engines of economic activity. Individuals borrow to build homes, educate their children and start businesses, while banks and microfinance institutions earn their income from the interest on those loans. Every loan application, however, carries risk. If an institution approves applicants who cannot repay, it suffers losses and may be forced to tighten credit for everyone; if it rejects applicants who would have repaid, deserving borrowers are shut out and the institution loses business. Deciding which applications to approve has traditionally relied on loan officers who manually review income, credit history, assets and the size of the requested loan. This process is slow, expensive and inconsistent, because two officers can reach different decisions on identical applications, and the delay can leave applicants waiting days or weeks for a response.

The problem this project addresses is therefore: *how can a financial institution decide quickly, consistently and objectively whether a loan application should be approved or rejected, using the information already provided by the applicant?* A data mining model is the most viable solution because banks hold large volumes of past applications whose outcomes are already known. A supervised classification algorithm can learn from these records which combinations of credit score, income, loan amount, loan term and asset values have historically led to approval, and then apply that learned pattern instantly to every new application. Such a model processes applications in milliseconds, applies the same rules to every applicant, and can be measured and audited with precise accuracy figures. It is intended to support loan officers by handling the routine cases and flagging uncertain ones, not to replace human judgement.

2. Methodology

Objective and category of model. The main objective is to build a model that classifies a loan application as *Approved* or *Rejected* with high accuracy, while keeping both false approvals and false rejections low. Because the outcome is one of two known labels, the task falls under **classification**, a category of supervised data mining. Classification was chosen over alternatives such as clustering or regression because the dataset contains labelled historical decisions and the required output is a category rather than a number. The dataset used is the Kaggle *Loan Approval Prediction Dataset* (loan_approval_dataset.csv), containing 4,269 applications and 13 columns: a

loan identifier, number of dependents, education, self-employment status, annual income, loan amount, loan term, CIBIL credit score, four asset values (residential, commercial, luxury and bank), and the target column *loan_status*. About 62.2% of the applications are approved and 37.8% rejected.

Steps and techniques. Data cleaning removed leading spaces from column names and text values, checked for missing values and duplicates (none were found), and corrected 28 impossible negative residential asset values by setting them to zero. Valid outliers in the asset columns were kept because they represent genuine wealthy applicants. The identifier column was dropped, education and self-employment were encoded as 0/1, and the target was encoded as Approved = 1 and Rejected = 0. Five ratio features commonly used by lenders were engineered: total assets, loan-to-income, loan-to-assets, monthly installment and installment-to-income. The data was split 80% training (3,415 rows) and 20% testing (854 rows) using stratified sampling and a fixed random seed of 42, so the class balance is preserved and results are reproducible. Four supervised algorithms were trained and compared: Logistic Regression, Decision Tree, Random Forest and Gradient Boosting, with the strongest candidates tuned using grid search. Validation was performed with 5-fold stratified cross-validation on the training data, a final evaluation on the untouched test set using accuracy, precision, recall, F1-score, ROC-AUC and a confusion matrix, and a 10-fold cross-validation on the full dataset to confirm stability.

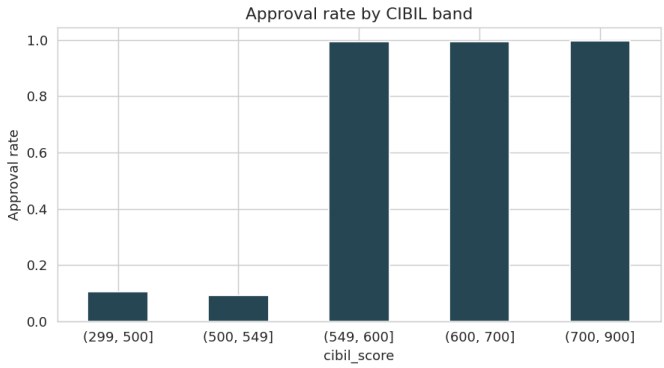


Figure 1: Approval rate by CIBIL score band. Applicants scoring 550 or above are approved about 99% of the time; those below are approved about 10% of the time.

3. Results and Discussion

During model comparison, 5-fold cross-validation on the training data gave the following accuracies: Decision Tree 99.97%, Gradient Boosting 99.91%, Random Forest 99.80% and Logistic Regression 97.16%. After hyperparameter tuning, Gradient Boosting (learning rate 0.05, depth 2, 200 trees) reached 99.97% and was selected as the final model because an ensemble is more stable than a single tree. On the held-out test set of 854 applications the final model achieved an **accuracy of 99.88%, precision of 99.81%, recall of 100%, F1-score of 99.91% and ROC-AUC of 1.00**. The confusion matrix shows 322 correctly rejected applications, 531 correctly approved applications, one rejected applicant wrongly approved and no approved applicant wrongly rejected. A 10-fold cross-validation on the full dataset gave 99.95% (standard deviation 0.09%), and diagnostic checks showed only a 0.12% gap between training and test accuracy, no change in accuracy when tree depth was increased, and stable results across 20 random splits, indicating the model neither overfits nor underfits on this dataset.

These scores show that the objective of building an accurate approval classifier has been met. Feature importance analysis revealed that the CIBIL credit score is by far the strongest driver of decisions, which explains why the simple Logistic Regression model scored lower: the relationship is a sharp cut-off at about 550 rather than a smooth trend. Several limitations should be noted. The near-perfect scores suggest the dataset is clean and strongly rule-driven, so real bank data, which is noisier, would likely produce lower accuracy. The single false approval is the costlier kind of error for a lender and should be monitored. Finally, features such as education and self-employment require a fairness review before any real-world use. The model is therefore a decision-support prototype and should be validated on current institutional data before deployment.

Model (5-fold CV, training data)	Accuracy	F1-score
Decision Tree	99.97%	99.98%
Gradient Boosting (tuned: final model)	99.97%	99.93%*
Random Forest	99.80%	99.84%
Logistic Regression	97.16%	97.70%

Table 1: Model comparison. *F1 shown is for the default (untuned) Gradient Boosting model.

Final model: test-set metric	Score
Accuracy	99.88%
Precision	99.81%
Recall	100.00%
F1-score	99.91%
ROC-AUC	1.000
10-fold CV accuracy (all data)	99.95% (+/- 0.09%)

Table 2: Performance of the final Gradient Boosting model on the 854-application test set.

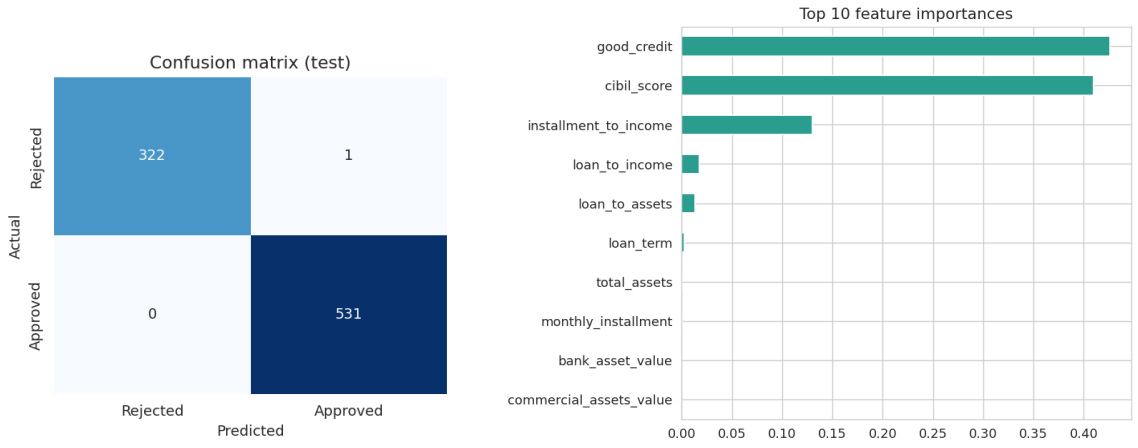


Figure 2: Confusion matrix (left) and top feature importances (right) of the final model.

4. Summary and Conclusion

This project applied supervised classification to predict loan approval using the Kaggle Loan Approval Prediction Dataset of 4,269 applications. The team cleaned the data, handled invalid

values, encoded categorical variables, engineered lender-style ratio features, split the data into training and testing sets, and compared Logistic Regression, Decision Tree, Random Forest and Gradient Boosting. The tuned Gradient Boosting model achieved 99.88% test accuracy, 99.81% precision, 100% recall and a 99.91% F1-score, and the validation checks confirmed that it generalises well without overfitting or underfitting, so the project objective was achieved. The credit score was found to be the dominant factor in approval decisions. To improve the model in future, we recommend testing it on larger and more realistic real-world lending data, adding features such as repayment history and employment duration, monitoring performance over time for data drift, conducting a fairness audit across applicant groups, adding explainability tools such as SHAP so decisions can be justified to applicants, and deploying it as a web application that supports, rather than replaces, human loan officers.

5. GitHub Repository

Project source code, documentation and model: **[INSERT YOUR GITHUB REPOSITORY LINK HERE]**